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Wisława Szymborska, “The poet and the world”, Nobel Lecture 1996. Polish original; English by Barańczak and Cavanagh. Copyright The Nobel Foundation 1996.

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Whatever inspiration is, it’s born from a continuous “I don’t know.”

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Każdym utworem próbuje na to odpowiedzieć, ale kiedy tylko postawi kropkę, już ogarnia go wahanie … Więc próbuje jeszcze raz i jeszcze raz.

Each poem marks an effort to answer this statement, but as soon as the final period hits the page, the poet begins to hesitate … So the poets keep on trying.

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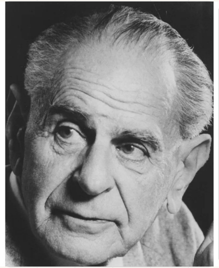
Popper, 1934: conjecture and refutation



Karl Popper, about 1980.

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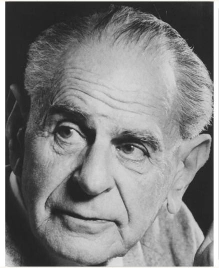

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Vienna, 1934. Popper asks what entitles anyone to believe a general rule drawn from finitely many cases.

His answer is that nothing does, and that this is survivable.

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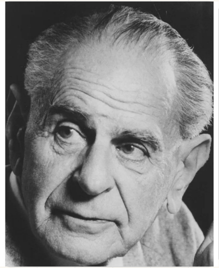

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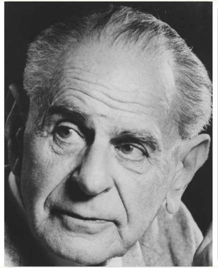

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We do not know: we can only guess.

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Popper’s tetradic schema

The way in which knowledge progresses … is by unjustified (and unjustifiable) anticipations, by guesses, by tentative solutions to our problems, by conjectures. These conjectures are controlled by criticism; that is, by attempted refutations …

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$$\boxed{P_1}$$

Problem

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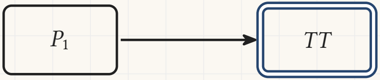

Problem   Tentative theory

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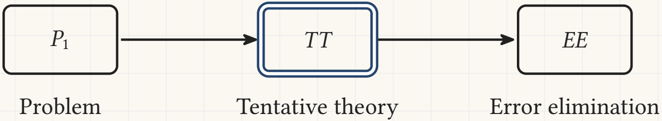

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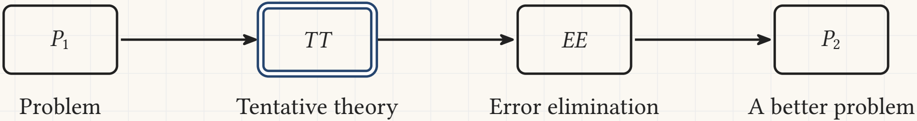

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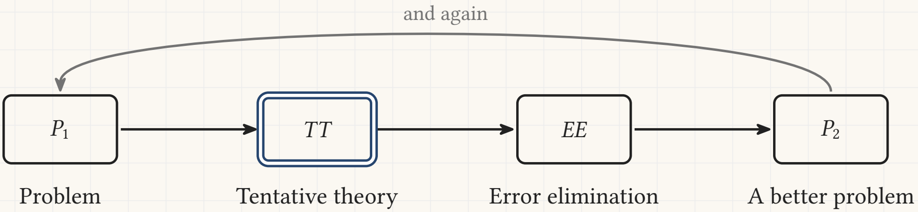

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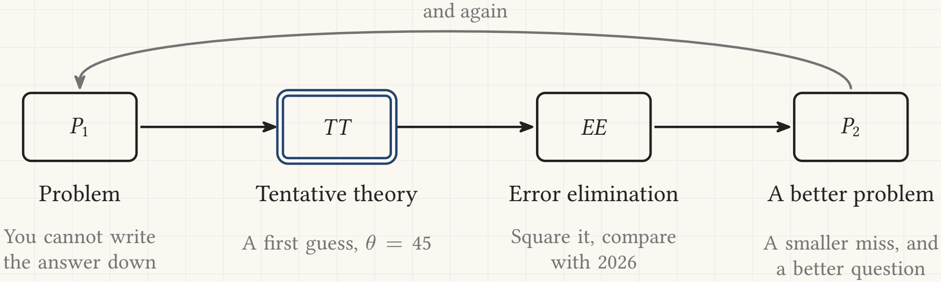

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Astronomy: the recovery of Ceres, 1801

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C. A. Jensen

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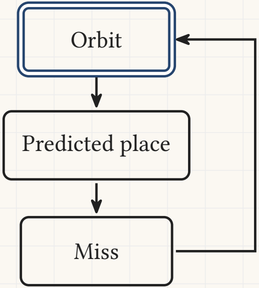

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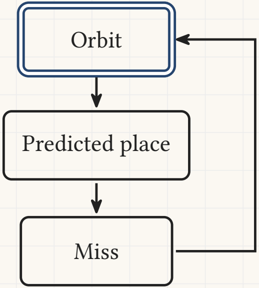
- Von Zach finds it again in December.



Computer science: Turing, 1950

Alan Turing, 29 March 1951.

Elliott and Fry

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The loop was written down as the plan for machine intelligence before there was a machine to run it on.

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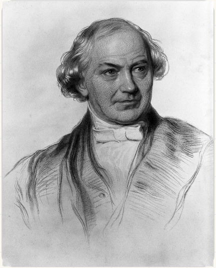
Whewell, 1858: trying wrong guesses



William Whewell, stipple engraving.

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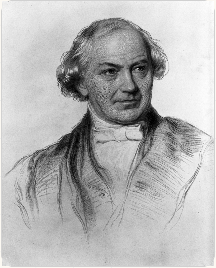

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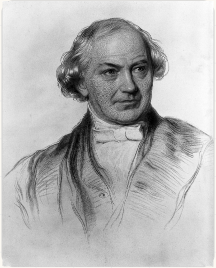

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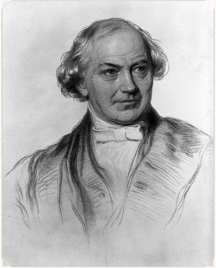

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Most guesses are wrong, and a step that aims too far is how the right one gets found.

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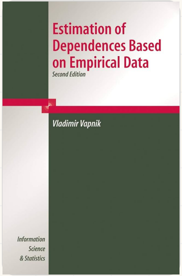
The loop has a name, and the name is a book



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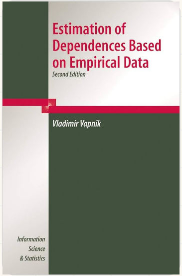

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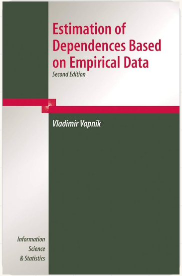

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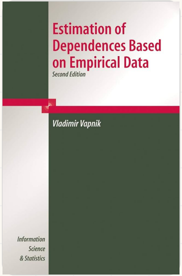

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The book sets out empirical risk minimisation, which is the strategy underneath most of what the rest of this course will build.

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The weekend

The weekend

Friday 4 September

08:00	Intro and onboarding
09:00	First-order approximation algorithms
10:00	Exercises: first-order optimization
10:30	*Coffee, Polysnack*
11:00	Gradient descent
12:00	Exercises: gradient descent
13:00	*Lunch, Dozentenfoyer*
14:00	Neural networks and universal approximation
14:30	Exercises: neural networks
15:30	*Coffee, Polysnack*

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Saturday 5 September

08:00	Training neural networks with PyTorch
09:00	Exercises: PyTorch
10:00	*Coffee, Cafebar*
10:30	Validation and overfitting
11:00	Exercises: validation and overfitting
12:00	Project intro: LLM routing

Every lecture is followed by its exercise session. The exercise slugs live on Code Expert and in the onboarding deck.

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16:00	**Vapnik's statistical learning theory**

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